

# Practical 5 - Non-proportional hazards and model assessment

## Survival Analysis

London School of Hygiene & Tropical Medicine

### SOLUTIONS

Datasets required: rotterdam (part of the survival package)

R packages required: survival, aod

```
library(survival)
library(aod)
```

### Introduction

In this practical we will use the methods introduced in the notes/lecture to assess some of the assumptions that underpin the Cox proportional hazards model. This practical uses the Rotterdam breast cancer study data, which contains information on 2982 individuals who were followed up after undergoing surgery for breast cancer. Several variables were collected on the participants at the start of follow-up. In this practical we will focus on the variables summarised in the table below. Our focus will be on the outcome of death. The models used in this practical are not designed to answer a specific question, but to help us practice using the techniques introduced in the lecture.

| Variable | Description                                              |
|----------|----------------------------------------------------------|
| pid      | Study ID number                                          |
| dtime    | Time to death or censoring in days                       |
| death    | Event indicator: 0=Censored, 1=Died                      |
| age      | Age at the start of follow-up, in years                  |
| hormon   | Indicator of receiving hormone therapy: 0=No, 1=Yes      |
| size     | Tumour size: 1: $\leq 20$ mm, 2: 20 – 50mm, 3: $> 50$ mm |

| Variable | Description |
|----------|-------------|
|----------|-------------|

The data is part of the survival package so does not need to be loaded separately.

## Aims

The aims of this session are:

- Investigate the proportional hazards assumption using covariate-by-time interactions.
- Investigate the proportional hazards assumption using Schoenfeld residuals.
- Fit and interpret extended Cox models that relax the proportional hazards assumption.
- Use Martingale residuals to investigate suitable functional forms for continuous covariates.

## Question 1

- Load the data, and create a new time variable that measures time in years instead of days.
- Summarise the survival data: how many events are there and what is the range of follow-up times.
- Obtain a Kaplan-Meier plot for overall survival, and interpret.

## SOLUTION

```
#data
head(rotterdam[,c("pid","mtime","death","age","hormon","size")])
```

```
      pid mtime death age hormon  size
1393   1  1799     0  74      0  <=20
1416   2  2828     0  79      0 20-50
2962   3  6012     0  44      0  <=20
1455   4  2624     0  70      0 20-50
977    5  4915     0  75      0  <=20
617    6  5888     0  52      0  <=20
```

```
#make time in years instead of days
rotterdam$mtime_yrs<-rotterdam$mtime/365.25

#number of events and follow-up duration
table(rotterdam$death)
```

```

0      1
1710 1272

```

```
summary(rotterdam$dttime_yrs)
```

```

      Min.   1st Qu.   Median     Mean  3rd Qu.     Max.
0.09856  4.39973   7.22382   7.13303  9.73237  19.28268

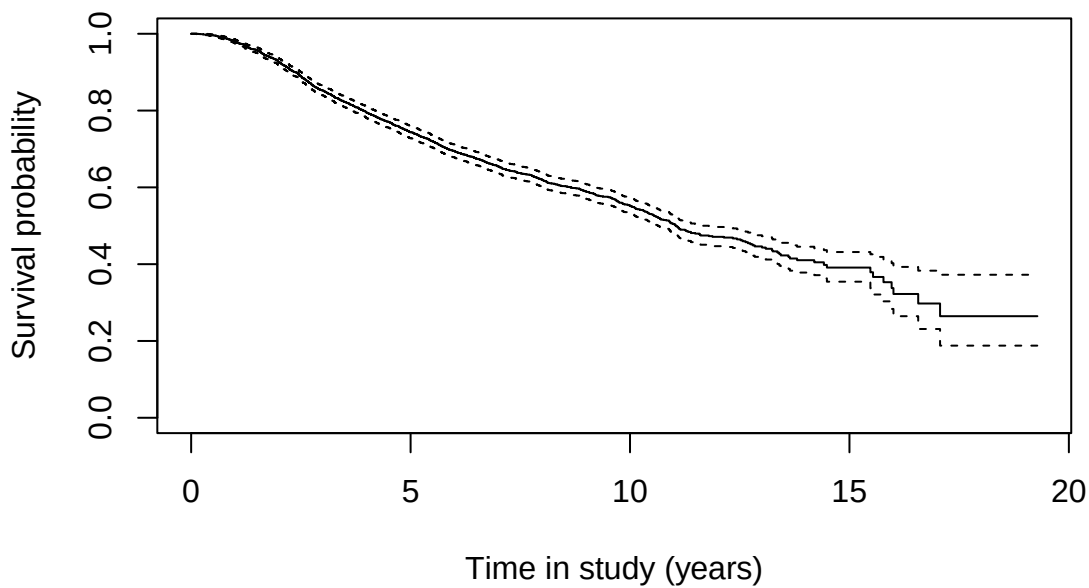
```

```

#Kaplan-Meier analysis
rdm.km<-survfit(Surv(dttime_yrs,death)~1,data=rotterdam)

plot(rdm.km,xlab="Time in study (years)",ylab="Survival probability")

```



```
summary(rdm.km,times=c(5,10,15))
```

```
Call: survfit(formula = Surv(dttime_yrs, death) ~ 1, data = rotterdam)
```

| time | n.risk | n.event | survival | std.err | lower 95% CI | upper 95% CI |
|------|--------|---------|----------|---------|--------------|--------------|
| 5    | 2084   | 753     | 0.744    | 0.00807 | 0.728        | 0.760        |
| 10   | 685    | 418     | 0.552    | 0.01040 | 0.532        | 0.573        |
| 15   | 43     | 94      | 0.391    | 0.01966 | 0.354        | 0.432        |

- Among the 2982 individuals there were 1272 deaths. Follow-up ranged from 36 days to 19.28 years.
- The Kaplan-Meier estimates of the probabilities of survival beyond 5, 10, and 15 years are respectively 74.4%, 55.2% and 39.1%.

## Question 2

- Fit a Cox proportional hazards model for the hazard of death, including the following explanatory variables: `age`, `hormon`, `size`.
- Interpret the hazard ratio estimates.

## SOLUTION

```
#Cox model
rdm.cox<-coxph(Surv(dtime_yrs,death)~age+hormon+as.factor(size),data=rotterdam)
summary(rdm.cox)
```

Call:

```
coxph(formula = Surv(dtime_yrs, death) ~ age + hormon + as.factor(size),
      data = rotterdam)
```

n= 2982, number of events= 1272

|                      | coef    | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|---------|-----------|----------|--------|--------------|
| age                  | 0.01411 | 1.01421   | 0.00225  | 6.271  | 3.59e-10 *** |
| hormon               | 0.15232 | 1.16454   | 0.08717  | 1.748  | 0.0806 .     |
| as.factor(size)20-50 | 0.64089 | 1.89817   | 0.06329  | 10.127 | < 2e-16 ***  |
| as.factor(size)>50   | 1.23151 | 3.42641   | 0.08620  | 14.287 | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 1.014     | 0.9860     | 1.0097    | 1.019     |
| hormon               | 1.165     | 0.8587     | 0.9817    | 1.381     |
| as.factor(size)20-50 | 1.898     | 0.5268     | 1.6767    | 2.149     |
| as.factor(size)>50   | 3.426     | 0.2919     | 2.8938    | 4.057     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 294.6 on 4 df, p=<2e-16

Wald test = 307.4 on 4 df, p=<2e-16

Score (logrank) test = 331.1 on 4 df, p=<2e-16

```
#joint Wald test for size (using aod package)
wald.test(b = coef(rdm.cox), Sigma = vcov(rdm.cox), Terms = c(3,4))
```

Wald test:

-----

Chi-squared test:

X2 = 219.9, df = 2, P(> X2) = 0.0

The results are summarised in the table.

- A 1 year increase in age is associated with a 1.4% increase in the hazard for death, conditional on hormone therapy and tumour size,. The 95% CI excludes 1 and the p-value is small. There is a strong association between age and the hazard, conditional on the other variables in the model.
- Hormone therapy use is associated with an increase of 16.5% in the hazard for death, conditional on age and tumour size, however the 95% CI includes 1, indicating no evidence of an association conditional on age and tumour size.
- Having a tumour size of 20 – 50mm is associated with an 89.8% increase in the hazard for death compared to those with tumour size  $\leq 20$ mm, conditional on age and hormone therapy use. Having a tumour size of  $> 50$ mm is associated with a 243% increase in the hazard for death compared to those with tumour size  $\leq 20$ mm, conditional on age and hormone therapy use. The joint Wald test for tumour size variables gives a p-value of  $< 0.001$ , indicating strong evidence of an association with the hazard.

| Variable            | HR            | 95% CI         |
|---------------------|---------------|----------------|
| Age (years)         | 1.014         | (1.010, 1.019) |
| Hormone therapy use | 1.165         | (0.982, 1.381) |
| Tumour size:        |               |                |
| $\leq 20$ mm        | 1 (reference) | -              |
| 20 – 50mm           | 1.898         | (1.677, 2.149) |
| $> 50$ mm           | 3.426         | (2.894, 4.057) |

### Question 3

- Using the code below, reformat the data in such a way that each person's follow-up is divided into the following time periods (in years) up to their end of follow-up due to death or censoring: (0, 5], (5, 10], (10, 15], (15, 20]. We will refer to this as the 'split data'.

```
#Split the data into time periods
split.times1<-seq(5,20,5)
rdm.split1<-survSplit(Surv(dtime_yrs,death)~.,cut=split.times1,
                      data=rotterdam,
                      ,start="start.time",end="stop.time",
                      event="death",episode="period")
```

- Take a look at the resulting data to understand how each person's follow-up is now formatted. How many rows are there in the split data?
- Refit the Cox model fitted in question 2 using the split data. To do this we need to take into account that each row in the data now represents a different period of follow-up, with a different start time and end time. This needs to be accounted for when we fit the model, through how we specify `Surv(time,time2,event)`.
- Check that you get the same results from this model as you got in question 2.

## SOLUTION

```
#Split the data into time periods
split.times1<-seq(5,20,5)
rdm.split1<-survSplit(Surv(dtime_yrs,death)~.,cut=split.times1,
                      data=rotterdam,
                      ,start="start.time",end="stop.time",
                      event="death",episode="period")

dim(rdm.split1)
```

```
[1] 5794    18
```

```
#refit the Cox model
rdm.cox.split1<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
                      age+hormon+as.factor(size),data=rdm.split1)
summary(rdm.cox.split1)
```

Call:

```
coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age + hormon + as.factor(size), data = rdm.split1)
```

```
n= 5794, number of events= 1272
```

|        | coef    | exp(coef) | se(coef) | z     | Pr(> z )     |
|--------|---------|-----------|----------|-------|--------------|
| age    | 0.01411 | 1.01421   | 0.00225  | 6.271 | 3.59e-10 *** |
| hormon | 0.15232 | 1.16454   | 0.08717  | 1.748 | 0.0806 .     |

```
as.factor(size)20-50 0.64089 1.89817 0.06329 10.127 < 2e-16 ***
as.factor(size)>50 1.23151 3.42641 0.08620 14.287 < 2e-16 ***
---
```

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 1.014     | 0.9860     | 1.0097    | 1.019     |
| hormon               | 1.165     | 0.8587     | 0.9817    | 1.381     |
| as.factor(size)20-50 | 1.898     | 0.5268     | 1.6767    | 2.149     |
| as.factor(size)>50   | 3.426     | 0.2919     | 2.8938    | 4.057     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 294.6 on 4 df, p=<2e-16

Wald test = 307.4 on 4 df, p=<2e-16

Score (logrank) test = 331.1 on 4 df, p=<2e-16

- There are 5794 rows in the split or 'long' data, compared to 2982 rows in the original data. Individuals can now have multiple rows in the data, representing their follow-up in up to four time periods (those in which they are at risk).

## Question 4

- Return to the original data. Reformat the data again in such a way that each person's follow-up is divided into small periods that are divided by the observed event times. Modify the code above to do this.
- Take a look at the resulting data to understand how each person's follow-up is now formatted. How many rows are there now in the split data?
- Refit the Cox model fitted in question 2 using the split data, using the same approach as used in question 3.
- Check that you get the same results from this model as you got in questions 2 and 3.

## SOLUTION

```
#Split the data based on the event times
split.times2<-rotterdam$dttime_yrs[rotterdam$death==1]
rdm.split2<-survSplit(Surv(dttime_yrs,death)~.,cut=split.times2,
                      data=rotterdam,
                      start="start.time",end="stop.time",
                      event="death",episode="period")
dim(rdm.split2)
```

```
[1] 2162531 18
```

```
#refit the Cox model
rdm.cox.split2<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
                      age+hormon+as.factor(size),data=rdm.split2)
summary(rdm.cox.split2)
```

Call:

```
coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age + hormon + as.factor(size), data = rdm.split2)
```

n= 2162531, number of events= 1272

|                      | coef    | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|---------|-----------|----------|--------|--------------|
| age                  | 0.01411 | 1.01421   | 0.00225  | 6.271  | 3.59e-10 *** |
| hormon               | 0.15232 | 1.16454   | 0.08717  | 1.748  | 0.0806 .     |
| as.factor(size)20-50 | 0.64089 | 1.89817   | 0.06329  | 10.127 | < 2e-16 ***  |
| as.factor(size)>50   | 1.23151 | 3.42641   | 0.08620  | 14.287 | < 2e-16 ***  |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 1.014     | 0.9860     | 1.0097    | 1.019     |
| hormon               | 1.165     | 0.8587     | 0.9817    | 1.381     |
| as.factor(size)20-50 | 1.898     | 0.5268     | 1.6767    | 2.149     |
| as.factor(size)>50   | 3.426     | 0.2919     | 2.8938    | 4.057     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 294.6 on 4 df, p=<2e-16

Wald test = 307.4 on 4 df, p=<2e-16

Score (logrank) test = 331.1 on 4 df, p=<2e-16

- There are now over 2 million rows in the data (2,162,531), because each individual has a row of data for every observed event time at which they appear in the risk set.

## Question 5

Using the split data created in question 4, we will assess the proportional hazards assumption for the `hormon` variable by adding an interaction between age and time in the Cox model. The 'time' variable that you use here should be `stop.time`, which is the end time of each small interval.

- Write down the form for the hazard model being fitted.



- Fit the model and interpret the results.
- Repeat for each variable included in the model, i.e. including an interaction between each variable and time.
- Assess the null hypothesis of proportionality of hazards for each variable.

## SOLUTION

```
#age
rdmtt.age<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
                  age+age:stop.time+hormon+as.factor(size),
                  data=rdm.split2)
summary(rdmtt.age)
```

Call:

```
coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age + age:stop.time + hormon + as.factor(size), data = rdm.split2)
```

n= 2162531, number of events= 1272

|                      | coef      | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|-----------|-----------|----------|--------|--------------|
| age                  | -0.000466 | 0.999534  | 0.004169 | -0.112 | 0.9110       |
| hormon               | 0.180369  | 1.197660  | 0.087484 | 2.062  | 0.0392 *     |
| as.factor(size)20-50 | 0.644443  | 1.904926  | 0.063281 | 10.184 | < 2e-16 ***  |
| as.factor(size)>50   | 1.233275  | 3.432451  | 0.086062 | 14.330 | < 2e-16 ***  |
| age:stop.time        | 0.003106  | 1.003111  | 0.000749 | 4.147  | 3.36e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 0.9995    | 1.0005     | 0.9914    | 1.008     |
| hormon               | 1.1977    | 0.8350     | 1.0089    | 1.422     |
| as.factor(size)20-50 | 1.9049    | 0.5250     | 1.6827    | 2.156     |
| as.factor(size)>50   | 3.4325    | 0.2913     | 2.8997    | 4.063     |
| age:stop.time        | 1.0031    | 0.9969     | 1.0016    | 1.005     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 311.9 on 5 df, p=<2e-16

Wald test = 325.6 on 5 df, p=<2e-16

Score (logrank) test = 350 on 5 df, p=<2e-16

```
#hormon
rdmtt.hormon<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
```

```

age+hormon+hormon:stop.time+as.factor(size),
data=rdm.split2)
summary(rdm.ttt.hormon)

```

Call:

```

coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age + hormon + hormon:stop.time + as.factor(size), data = rdm.split2)

```

n= 2162531, number of events= 1272

|                      | coef     | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|----------|-----------|----------|--------|--------------|
| age                  | 0.014166 | 1.014267  | 0.002251 | 6.293  | 3.11e-10 *** |
| hormon               | 0.073896 | 1.076695  | 0.163518 | 0.452  | 0.651        |
| as.factor(size)20-50 | 0.640111 | 1.896692  | 0.063303 | 10.112 | < 2e-16 ***  |
| as.factor(size)>50   | 1.232235 | 3.428885  | 0.086187 | 14.297 | < 2e-16 ***  |
| hormon:stop.time     | 0.019455 | 1.019645  | 0.033927 | 0.573  | 0.566        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 1.014     | 0.9859     | 1.0098    | 1.019     |
| hormon               | 1.077     | 0.9288     | 0.7815    | 1.483     |
| as.factor(size)20-50 | 1.897     | 0.5272     | 1.6754    | 2.147     |
| as.factor(size)>50   | 3.429     | 0.2916     | 2.8959    | 4.060     |
| hormon:stop.time     | 1.020     | 0.9807     | 0.9540    | 1.090     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 294.9 on 5 df, p=<2e-16

Wald test = 307.8 on 5 df, p=<2e-16

Score (logrank) test = 331.6 on 5 df, p=<2e-16

```

#size
#Note that it is helpful to create two separate dummy variables
# for the size variables here
rdm.split2$size_2050<-ifelse(rdm.split2$size=="20-50",1,0)
rdm.split2$size_50plus<-ifelse(rdm.split2$size==">50",1,0)
rdm.ttt.size<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
                    age+hormon+size_2050+stop.time:size_2050+
                    size_50plus+stop.time:size_50plus,data=rdm.split2)
summary(rdm.ttt.size)

```

Call:

```
coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age + hormon + size_2050 + stop.time:size_2050 + size_50plus +
      stop.time:size_50plus, data = rdm.split2)
```

```
n= 2162531, number of events= 1272
```

|                       | coef     | exp(coef) | se(coef) | z      | Pr(> z )     |
|-----------------------|----------|-----------|----------|--------|--------------|
| age                   | 0.01397  | 1.01407   | 0.00225  | 6.210  | 5.31e-10 *** |
| hormon                | 0.14518  | 1.15625   | 0.08731  | 1.663  | 0.09636 .    |
| size_2050             | 0.91405  | 2.49442   | 0.12225  | 7.477  | 7.59e-14 *** |
| size_50plus           | 1.55390  | 4.72990   | 0.16070  | 9.670  | < 2e-16 ***  |
| size_2050:stop.time   | -0.05288 | 0.94849   | 0.02015  | -2.624 | 0.00868 **   |
| stop.time:size_50plus | -0.06703 | 0.93517   | 0.03020  | -2.220 | 0.02645 *    |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

|                       | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-----------------------|-----------|------------|-----------|-----------|
| age                   | 1.0141    | 0.9861     | 1.0096    | 1.0186    |
| hormon                | 1.1563    | 0.8649     | 0.9744    | 1.3721    |
| size_2050             | 2.4944    | 0.4009     | 1.9630    | 3.1697    |
| size_50plus           | 4.7299    | 0.2114     | 3.4520    | 6.4809    |
| size_2050:stop.time   | 0.9485    | 1.0543     | 0.9118    | 0.9867    |
| stop.time:size_50plus | 0.9352    | 1.0693     | 0.8814    | 0.9922    |

```
Concordance= 0.637 (se = 0.008 )
```

```
Likelihood ratio test= 303.1 on 6 df, p=<2e-16
```

```
Wald test = 309.5 on 6 df, p=<2e-16
```

```
Score (logrank) test = 337.1 on 6 df, p=<2e-16
```

- The model is  $h(t|age, hormon, size) = h_0(t) \exp\{\beta_1 age + \gamma_1 age \times t + \beta_2 hormon + \beta_3 size_{20to50} + \beta_4 size_{50plus}\}$  where  $size_{20to50}$  takes value 1 if tumour size is in the range 20 – 50mm and 0 otherwise, and  $size_{50plus}$  takes value 1 if tumour size is in the range > 50mm and 0 otherwise.
- The parameter  $\gamma_1$  is the coefficient for the  $age \times t$  interaction term. The HR estimate is  $e^{\hat{\gamma}_1} = 1.003$ , with 95% CI (1.002, 1.005), and the p-value for a Wald test of the null hypothesis that  $\gamma_1 = 0$  is very small ( $< 0.0001$ ). There is strong evidence that the proportional hazards assumption is not valid for age. The estimate of  $\gamma_1$  is greater than 0, telling us that the hazard ratio for age increases with time.
- There is no evidence against the proportional hazards assumption for the hormone therapy variable. There is evidence against the proportional hazards assumption for both tumour size variables. The HRs for the interaction terms are less than 1 (0.949 and 0.935), suggesting that the impact of tumour size on the hazard reduces over time.

## Question 6

Return to the original data. There is a ready-made function that allows you to fit the models in question 5, with covariate-by-time interactions, without having to split the data on all event times first (the function does this for you in the background). Try out the code below and see if you get the same results as when you did this manually in question 5.

```
rdmtt.age.2<-coxph(Surv(dtime_yrs,death)~age+hormon+as.factor(size)+tt(age),
                    data=rotterdam,
                    tt=function(x,t,...){x*t})
summary(rdmtt.age.2)
```

## SOLUTION

```
#using tt option in coxph

#age
rdmtt.age.2<-coxph(Surv(dtime_yrs,death)~age+hormon+as.factor(size)+tt(age),
                    data=rotterdam,
                    tt=function(x,t,...){x*t})
summary(rdmtt.age.2)
```

Call:

```
coxph(formula = Surv(dtime_yrs, death) ~ age + hormon + as.factor(size) +
      tt(age), data = rotterdam, tt = function(x, t, ...) {
        x * t
      })
```

n= 2982, number of events= 1272

|                      | coef      | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|-----------|-----------|----------|--------|--------------|
| age                  | -0.000466 | 0.999534  | 0.004169 | -0.112 | 0.9110       |
| hormon               | 0.180369  | 1.197660  | 0.087484 | 2.062  | 0.0392 *     |
| as.factor(size)20-50 | 0.644443  | 1.904926  | 0.063281 | 10.184 | < 2e-16 ***  |
| as.factor(size)>50   | 1.233275  | 3.432451  | 0.086062 | 14.330 | < 2e-16 ***  |
| tt(age)              | 0.003106  | 1.003111  | 0.000749 | 4.147  | 3.36e-05 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|        | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|--------|-----------|------------|-----------|-----------|
| age    | 0.9995    | 1.0005     | 0.9914    | 1.008     |
| hormon | 1.1977    | 0.8350     | 1.0089    | 1.422     |

|                      |        |        |        |       |
|----------------------|--------|--------|--------|-------|
| as.factor(size)20-50 | 1.9049 | 0.5250 | 1.6827 | 2.156 |
| as.factor(size)>50   | 3.4325 | 0.2913 | 2.8997 | 4.063 |
| tt(age)              | 1.0031 | 0.9969 | 1.0016 | 1.005 |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 311.9 on 5 df, p=<2e-16

Wald test = 325.6 on 5 df, p=<2e-16

Score (logrank) test = 350 on 5 df, p=<2e-16

```
#hormon
rdmtt.hormon.2<-coxph(Surv(dtime_yrs,death)~age+hormon+
  as.factor(size)+tt(hormon),
  data=rotterdam,
  tt=function(x,t,...){x*t})
summary(rdmtt.hormon.2)
```

Call:

```
coxph(formula = Surv(dtime_yrs, death) ~ age + hormon + as.factor(size) +
  tt(hormon), data = rotterdam, tt = function(x, t, ...) {
  x * t
})
```

n= 2982, number of events= 1272

|                      | coef     | exp(coef) | se(coef) | z      | Pr(> z )     |
|----------------------|----------|-----------|----------|--------|--------------|
| age                  | 0.014166 | 1.014267  | 0.002251 | 6.293  | 3.11e-10 *** |
| hormon               | 0.073896 | 1.076695  | 0.163518 | 0.452  | 0.651        |
| as.factor(size)20-50 | 0.640111 | 1.896692  | 0.063303 | 10.112 | < 2e-16 ***  |
| as.factor(size)>50   | 1.232235 | 3.428885  | 0.086187 | 14.297 | < 2e-16 ***  |
| tt(hormon)           | 0.019455 | 1.019645  | 0.033927 | 0.573  | 0.566        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                      | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|----------------------|-----------|------------|-----------|-----------|
| age                  | 1.014     | 0.9859     | 1.0098    | 1.019     |
| hormon               | 1.077     | 0.9288     | 0.7815    | 1.483     |
| as.factor(size)20-50 | 1.897     | 0.5272     | 1.6754    | 2.147     |
| as.factor(size)>50   | 3.429     | 0.2916     | 2.8959    | 4.060     |
| tt(hormon)           | 1.020     | 0.9807     | 0.9540    | 1.090     |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 294.9 on 5 df, p=<2e-16

Wald test = 307.8 on 5 df, p=<2e-16  
 Score (logrank) test = 331.6 on 5 df, p=<2e-16

```
#size
rotterdam$size_2050<-ifelse(rotterdam$size=="20-50",1,0)
rotterdam$size_50plus<-ifelse(rotterdam$size==">50",1,0)
rdmtt.size.2<-coxph(Surv(dtime_yrs,death)~age+hormon+size_2050+size_50plus+
                    tt(size_2050)+tt(size_50plus),
                    data=rotterdam,
                    tt=function(x,t,...){x*t})
summary(rdmtt.size.2)
```

Call:

```
coxph(formula = Surv(dtime_yrs, death) ~ age + hormon + size_2050 +
      size_50plus + tt(size_2050) + tt(size_50plus), data = rotterdam,
      tt = function(x, t, ...) {
        x * t
      })
```

n= 2982, number of events= 1272

|                 | coef     | exp(coef) | se(coef) | z      | Pr(> z ) |     |
|-----------------|----------|-----------|----------|--------|----------|-----|
| age             | 0.01397  | 1.01407   | 0.00225  | 6.210  | 5.31e-10 | *** |
| hormon          | 0.14518  | 1.15625   | 0.08731  | 1.663  | 0.09636  | .   |
| size_2050       | 0.91405  | 2.49442   | 0.12225  | 7.477  | 7.59e-14 | *** |
| size_50plus     | 1.55390  | 4.72990   | 0.16070  | 9.670  | < 2e-16  | *** |
| tt(size_2050)   | -0.05288 | 0.94849   | 0.02015  | -2.624 | 0.00868  | **  |
| tt(size_50plus) | -0.06703 | 0.93517   | 0.03020  | -2.220 | 0.02645  | *   |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                 | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|-----------------|-----------|------------|-----------|-----------|
| age             | 1.0141    | 0.9861     | 1.0096    | 1.0186    |
| hormon          | 1.1563    | 0.8649     | 0.9744    | 1.3721    |
| size_2050       | 2.4944    | 0.4009     | 1.9630    | 3.1697    |
| size_50plus     | 4.7299    | 0.2114     | 3.4520    | 6.4809    |
| tt(size_2050)   | 0.9485    | 1.0543     | 0.9118    | 0.9867    |
| tt(size_50plus) | 0.9352    | 1.0693     | 0.8814    | 0.9922    |

Concordance= 0.637 (se = 0.008 )

Likelihood ratio test= 303.1 on 6 df, p=<2e-16

Wald test = 309.5 on 6 df, p=<2e-16

Score (logrank) test = 337.1 on 6 df,  $p < 2e-16$

You should get the same estimates as in question 5.

## Question 7

### Use the original data (i.e. not split on time)

In this question we will look at the Schoenfeld residuals for assessing the proportional hazards assumption.

- Based on the Cox model you already fitted in question 2, plot the Schoenfeld residuals for each variable using the code below. Assess the results.
- Compare your findings with those from question 5 (and 6).

```
#Cox model
rotterdam$size_2050<-ifelse(rotterdam$size=="20-50",1,0)
rotterdam$size_50plus<-ifelse(rotterdam$size==">50",1,0)
rdm.cox<-coxph(Surv(dtime_yrs,death)~age+hormon+size_2050+size_50plus,
               data=rotterdam)

#Schoenfeld residuals
sch.resid<-cox.zph(rdm.cox, transform = 'identity')
sch.resid

par(mfrow=c(2,2))
plot(sch.resid,col="red",lwd=2)
```

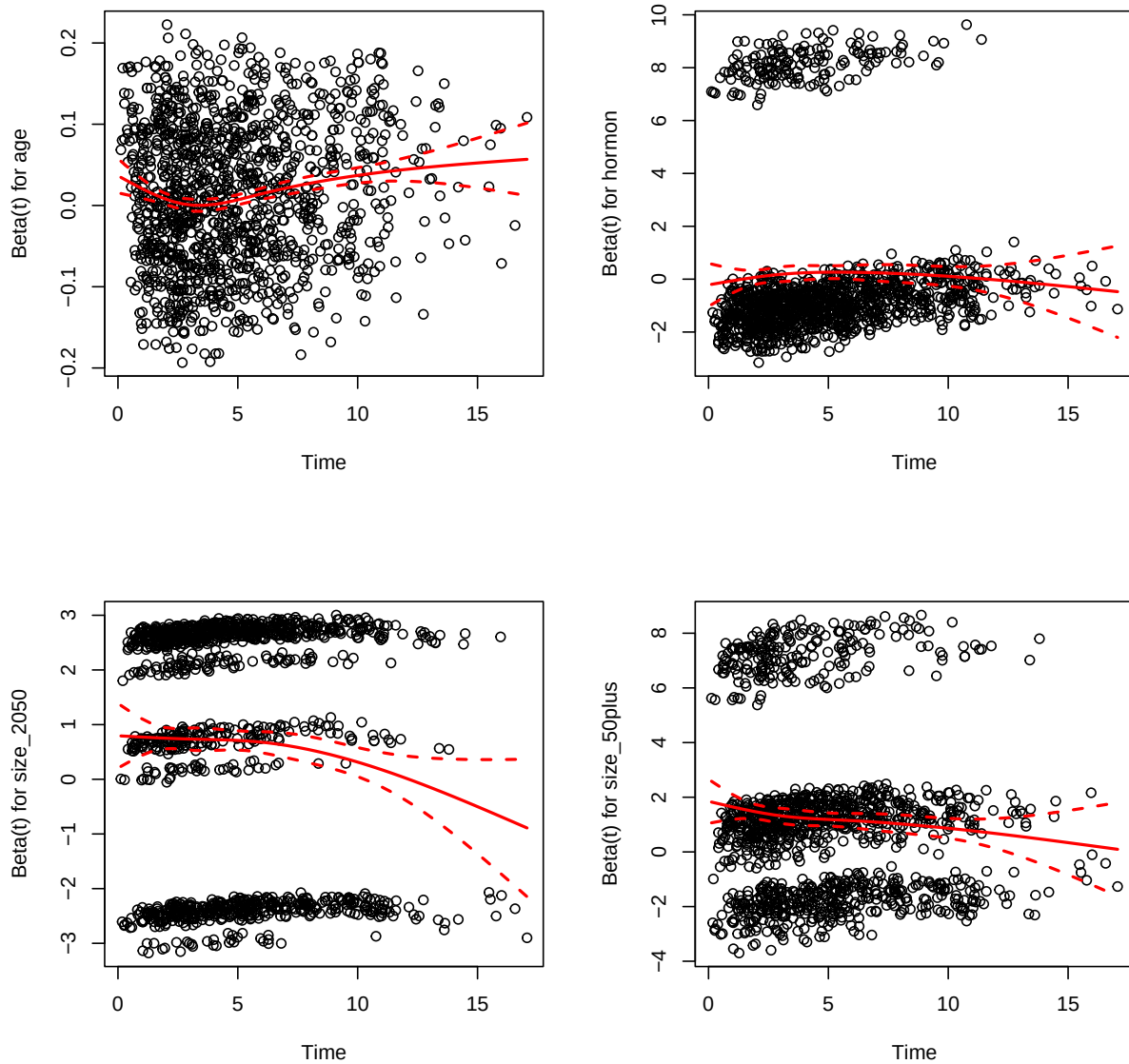
## SOLUTION

```
#Cox model
rotterdam$size_2050<-ifelse(rotterdam$size=="20-50",1,0)
rotterdam$size_50plus<-ifelse(rotterdam$size==">50",1,0)
rdm.cox<-coxph(Surv(dtime_yrs,death)~age+hormon+size_2050+size_50plus,
               data=rotterdam)

#Schoenfeld residuals
sch.resid<-cox.zph(rdm.cox, transform = 'identity')
sch.resid
```

|             | chisq  | df | p       |
|-------------|--------|----|---------|
| age         | 17.286 | 1  | 3.2e-05 |
| hormon      | 0.329  | 1  | 0.566   |
| size_2050   | 3.450  | 1  | 0.063   |
| size_50plus | 1.613  | 1  | 0.204   |
| GLOBAL      | 28.139 | 4  | 1.2e-05 |

```
par(mfrow=c(2,2))
plot(sch.resid,col="red",lwd=2)
```



- The Schoenfeld residual plot for age indicates an association between the residuals and



time. The shape of the smoothed curve suggests that the log HR for age is higher in the initial period of follow-up (up to around 2 years), before going down to around 0 and then curving slowly upwards for the remaining followup time.

- The Schoenfeld residual plot does not show any association between the residuals for hormone therapy and time. The smoothed line is around 0, which is consistent with the log hazard ratio being close to 0 and with its confidence interval including 0.
- The Schoenfeld residuals plots for the two tumour size variables show an association between the residuals and time. The smoothed curves are sloping down over time, and the curve even crosses zero for the *size20to50* level, indicating that the log hazard ratio goes from above zero at earlier times to below zero at later times.
- The results from the tests of association between the residuals and time correspond to the results from the tests of the variable-by-time interaction terms in question 5. The p-values are identical.

## Question 8

You may have found some evidence that the proportional hazards assumption may not be appropriate for **age** and **size**. In this question we will use an extended Cox model to relax the proportional hazards assumption for **age**.

- Go back to the split data that you created in question 3, where each person's follow-up was divided into 5 year periods. Fit an extended Cox model, allowing for an interaction between time period and age.
- What are the estimated hazard ratios for age in each of the four time periods?

## SOLUTION

```
#Cox model allowing the HR for age to vary over time
rdm.age.period<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
  age+age:as.factor(period)+hormon+as.factor(size),
  data=rdm.split1)
summary(rdm.age.period)
```

Call:

```
coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
  age + age:as.factor(period) + hormon + as.factor(size), data = rdm.split1)
```

n= 5794, number of events= 1272

|        | coef     | exp(coef) | se(coef) | z     | Pr(> z ) |    |
|--------|----------|-----------|----------|-------|----------|----|
| age    | 0.007569 | 1.007598  | 0.002856 | 2.650 | 0.00805  | ** |
| hormon | 0.184895 | 1.203092  | 0.087526 | 2.112 | 0.03465  | *  |

```

as.factor(size)20-50    0.647306   1.910387 0.063305 10.225 < 2e-16 ***
as.factor(size)>50      1.233702   3.433919 0.086080 14.332 < 2e-16 ***
age:as.factor(period)2 0.011194   1.011257 0.004814  2.325 0.02006 *
age:as.factor(period)3 0.044034   1.045018 0.009495  4.638 3.52e-06 ***
age:as.factor(period)4 0.041826   1.042713 0.038418  1.089 0.27628

```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|                        | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|------------------------|-----------|------------|-----------|-----------|
| age                    | 1.008     | 0.9925     | 1.0020    | 1.013     |
| hormon                 | 1.203     | 0.8312     | 1.0134    | 1.428     |
| as.factor(size)20-50   | 1.910     | 0.5235     | 1.6875    | 2.163     |
| as.factor(size)>50     | 3.434     | 0.2912     | 2.9008    | 4.065     |
| age:as.factor(period)2 | 1.011     | 0.9889     | 1.0018    | 1.021     |
| age:as.factor(period)3 | 1.045     | 0.9569     | 1.0257    | 1.065     |
| age:as.factor(period)4 | 1.043     | 0.9590     | 0.9671    | 1.124     |

Concordance= 0.638 (se = 0.008 )

Likelihood ratio test= 319.8 on 7 df, p=<2e-16

Wald test = 332.6 on 7 df, p=<2e-16

Score (logrank) test = 358.3 on 7 df, p=<2e-16

```

#another way of doing this
rdm.split1$age.p1<-ifelse(rdm.split1$period==1,rdm.split1$age,0)
rdm.split1$age.p2<-ifelse(rdm.split1$period==2,rdm.split1$age,0)
rdm.split1$age.p3<-ifelse(rdm.split1$period==3,rdm.split1$age,0)
rdm.split1$age.p4<-ifelse(rdm.split1$period==4,rdm.split1$age,0)
rdm.age.period2<-coxph(Surv(time=start.time,time2=stop.time,event=death)~
                        age.p1+age.p2+age.p3+age.p4+
                        hormon+as.factor(size),data=rdm.split1)
summary(rdm.age.period2)

```

Call:

```

coxph(formula = Surv(time = start.time, time2 = stop.time, event = death) ~
      age.p1 + age.p2 + age.p3 + age.p4 + hormon + as.factor(size),
      data = rdm.split1)

```

n= 5794, number of events= 1272

|        | coef     | exp(coef) | se(coef) | z     | Pr(> z )     |
|--------|----------|-----------|----------|-------|--------------|
| age.p1 | 0.007569 | 1.007598  | 0.002856 | 2.650 | 0.00805 **   |
| age.p2 | 0.018764 | 1.018941  | 0.003932 | 4.772 | 1.83e-06 *** |

```

age.p3          0.051603  1.052958 0.009061  5.695 1.23e-08 ***
age.p4          0.049395  1.050636 0.038302  1.290  0.19718
hormon          0.184895  1.203092 0.087526  2.112  0.03465 *
as.factor(size)20-50 0.647306  1.910387 0.063305 10.225 < 2e-16 ***
as.factor(size)>50  1.233702  3.433919 0.086080 14.332 < 2e-16 ***

```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```

              exp(coef) exp(-coef) lower .95 upper .95
age.p1          1.008      0.9925      1.0020      1.013
age.p2          1.019      0.9814      1.0111      1.027
age.p3          1.053      0.9497      1.0344      1.072
age.p4          1.051      0.9518      0.9747      1.133
hormon          1.203      0.8312      1.0134      1.428
as.factor(size)20-50  1.910      0.5235      1.6875      2.163
as.factor(size)>50    3.434      0.2912      2.9008      4.065

```

Concordance= 0.638 (se = 0.008 )

Likelihood ratio test= 319.8 on 7 df, p=<2e-16

Wald test = 332.6 on 7 df, p=<2e-16

Score (logrank) test = 358.3 on 7 df, p=<2e-16

- The table below shows the estimated hazard ratios for a 1 year increase in age in the four time periods, conditional on hormone therapy and tumour size. The hazard ratio for age is higher later in the course of follow-up, and is similar after 10 years.

| Time period   | HR    | 95% CI         |
|---------------|-------|----------------|
| ≤ 5 years     | 1.008 | (1.002,1.013)  |
| 5 – 10 years  | 1.019 | (1.011, 1.027) |
| 10 – 15 years | 1.053 | (1.034, 1.072) |
| > 15 years    | 1.051 | (0.975, 1.133) |

## Question 9

Use the original data (i.e. not split on time)

In this question we will relax the proportional hazards assumption for **size** by using a stratified Cox model.

- Write down the form of the Cox hazard model, assuming proportional hazards for **age** and **hormon** and a separate baseline hazard for each category of the size variable.
- Fit the model using the **strata()** option in **coxph**.

- Interpret the results.

## SOLUTION

```
rdm.size.strat<-coxph(Surv(time=dtime_yrs,event=death)~age+hormon+
                      strata(size),data=rotterdam)
summary(rdm.size.strat)
```

Call:

```
coxph(formula = Surv(time = dtime_yrs, event = death) ~ age +
      hormon + strata(size), data = rotterdam)
```

n= 2982, number of events= 1272

|        | coef     | exp(coef) | se(coef) | z     | Pr(> z )    |
|--------|----------|-----------|----------|-------|-------------|
| age    | 0.013851 | 1.013947  | 0.002251 | 6.153 | 7.6e-10 *** |
| hormon | 0.147265 | 1.158661  | 0.087353 | 1.686 | 0.0918 .    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

|        | exp(coef) | exp(-coef) | lower .95 | upper .95 |
|--------|-----------|------------|-----------|-----------|
| age    | 1.014     | 0.9862     | 1.0095    | 1.018     |
| hormon | 1.159     | 0.8631     | 0.9763    | 1.375     |

Concordance= 0.541 (se = 0.009 )

Likelihood ratio test= 45.18 on 2 df, p=2e-10

Wald test = 45.09 on 2 df, p=2e-10

Score (logrank) test = 45.35 on 2 df, p=1e-10

- The form of the model is  $h(t|age, hormon, size = s) = h_{0s}(t) \exp\{\beta_1 age + \beta_2 hormon\}$ , where  $s$  denotes tumour size group.
- We see in the results that there are no longer any hazard ratios for the size variable. The hazard ratios for age and hormone therapy are similar to those seen in the original model, where we also assumed proportional hazards for tumour size. The hazard ratios for age and hormone therapy are still conditional on tumour size – it is just that we have now conditioned on tumour size by allowing different baseline hazards by size group, instead of by including it as a regression term and making the proportional hazards assumption.

## Question 10

Use the original (i.e. not split on time) data

In this question we will use Martingale residuals to investigate whether the functional form for our continuous variable `age` is appropriate, or whether transforming age might provide a better fit.

- Fit a Cox proportional hazards model including `hormon` and `size`, but *excluding* `age`. Use the code below to obtain the Martingale residuals from the model and plot them against age. What is the association between the residuals and age and what does this suggest to you about what an appropriate functional form might be for age in the model?

```
#Martingale residual plot for age: excluding age from Cox model
rdm.cox.noage<-coxph(Surv(time=dtime_yrs,event=death)~
                    hormon+as.factor(size),data=rotterdam)
mgale.res.noage<-resid(rdm.cox.noage,type="martingale")

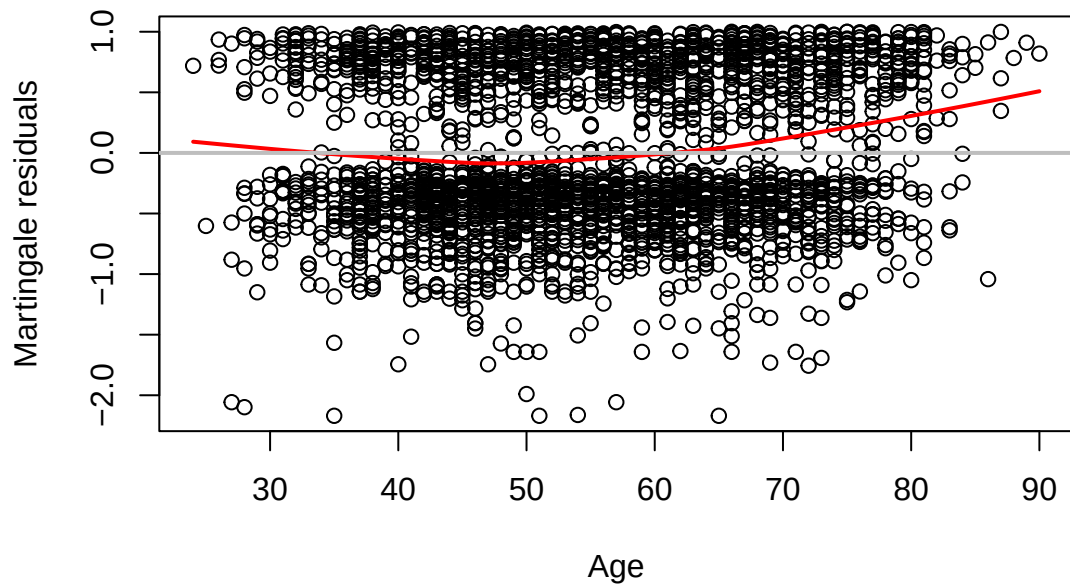
plot(rotterdam$age,mgale.res.noage,
     xlab="Age",ylab="Martingale residuals")
lines(lowess(rotterdam$age,mgale.res.noage),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```

- Fit a Cox proportional hazards model including `age`, `hormon` and `size`. Obtain the Martingale residuals from the model and plot them against age. What is the association between the residuals and age and what does this suggest to you about what an appropriate functional form might be for age in the model?
- If you think a transformation of age might improve the model fit, do this and reassess the model fit.

## SOLUTION

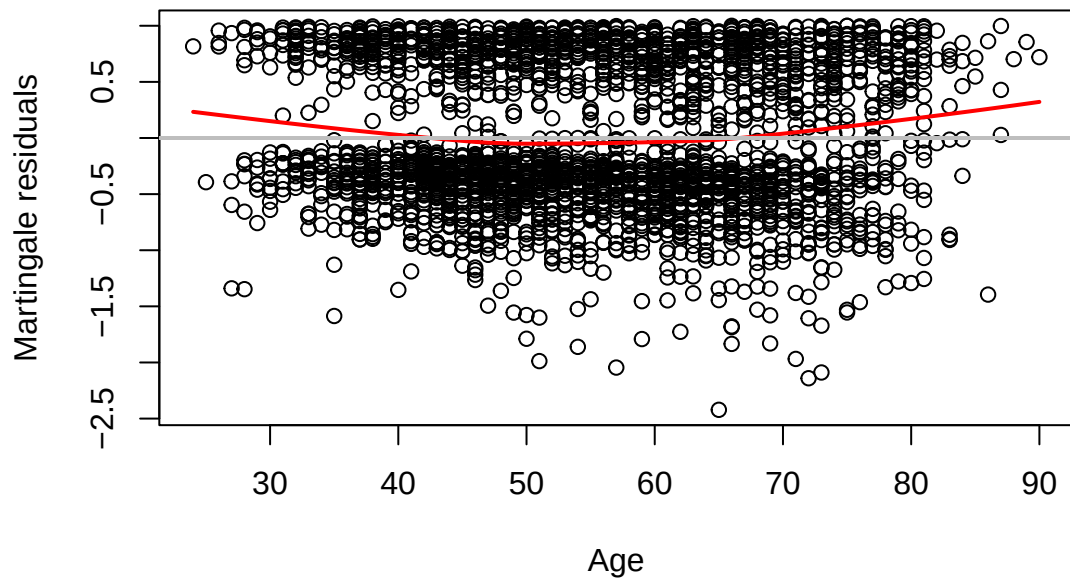
```
#Martingale residual plot for age: excluding age from Cox model
rdm.cox.noage<-coxph(Surv(time=dtime_yrs,event=death)~hormon+as.factor(size),
                    data=rotterdam)
mgale.res.noage<-resid(rdm.cox.noage,type="martingale")

plot(rotterdam$age,mgale.res.noage,
     xlab="Age",ylab="Martingale residuals")
lines(lowess(rotterdam$age,mgale.res.noage),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```



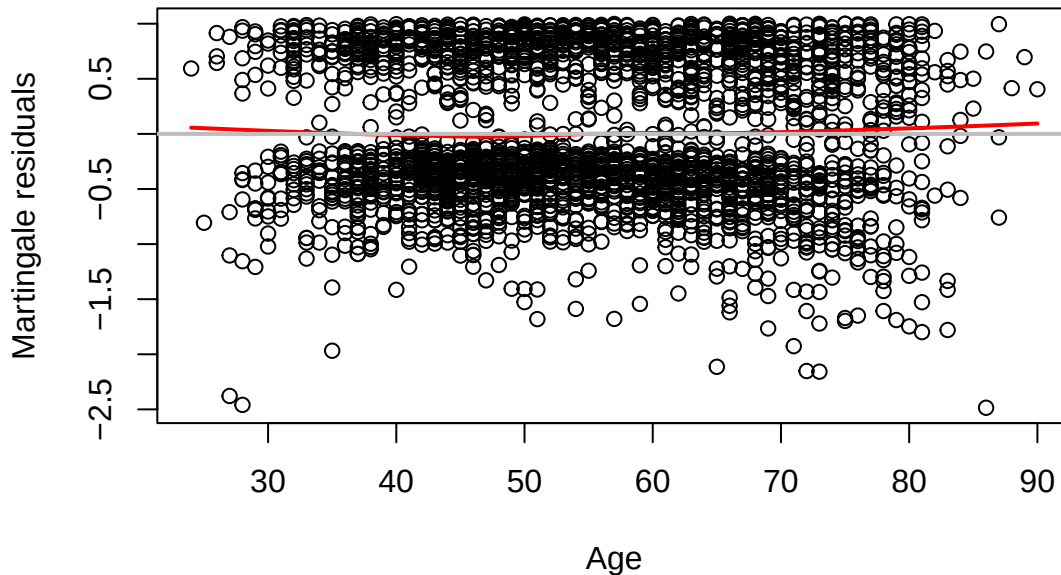
```
#Martingale residual plot for age: including age in Cox model
rdm.cox.age<-coxph(Surv(time=dtime_yrs,event=death)~age+hormon+as.factor(size),
                    data=rotterdam)
mgale.res.age<-resid(rdm.cox.age,type="martingale")

plot(rotterdam$age,mgale.res.age,xlab="Age",ylab="Martingale residuals")
lines(lowess(rotterdam$age,mgale.res.age),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```



```
#Martingale residual plot for age: including age and age^2 in Cox model
rdm.cox.age2<-coxph(Surv(time=dtime_yrs,event=death)~age+I(age^2)+
                    hormon+as.factor(size),data=rotterdam)
mgale.res.age2<-resid(rdm.cox.age2,type="martingale")

plot(rotterdam$age,mgale.res.age2,ylab="Martingale residuals",xlab="Age")
lines(lowess(rotterdam$age,mgale.res.age2),col="red",lwd=2)
abline(h=0,lwd=2,col="grey")
```



- There is a curved, perhaps approximately quadratic, association between the Martingale residuals and age. This suggests that the association between age and log hazard is not linear.
- There remains an association between the Martingale residuals and age even after we have included age in the model using a linear term. This indicates that the model has not completely captured the association between age and the hazard.
- We added an  $age^2$  term into the hazard model and reassessed the Martingale residuals. A plot of the Martingale residuals based on this model against age shows little association between the residuals and age. This suggests we have improved the fit of the model by adding the  $age^2$  term, and there is little residual relationship between age and the hazard.